

SMART BERTH MONITORING • MMWAVE • BLE MESH • MQTT/MTLS

Dock*Pulse*

Sensor-verified berth occupancy for small and medium marinas.

TEAM

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DEPLOYED

dockpulse.xyz

Berth availability is *guesswork*, not data.

01

TODAY

Manual flags.

Green means leaving, red means staying. Staff walk the dock to find out.

02

TODAY

Software platforms.

Tenants tap a button to declare a berth free. Records drift the moment someone forgets.

03

TODAY

Enterprise sensors.

Closed, sensor-verified platforms exist — priced for commercial ports, not local harbors.

One radar. One gateway. *One dashboard.*

01

Radar.

1 PER BERTH

24 GHz mmWave · solar · BLE
Mesh

02

Gateway.

1 PER HARBOR

ESP32-C3 · MQTT over mTLS

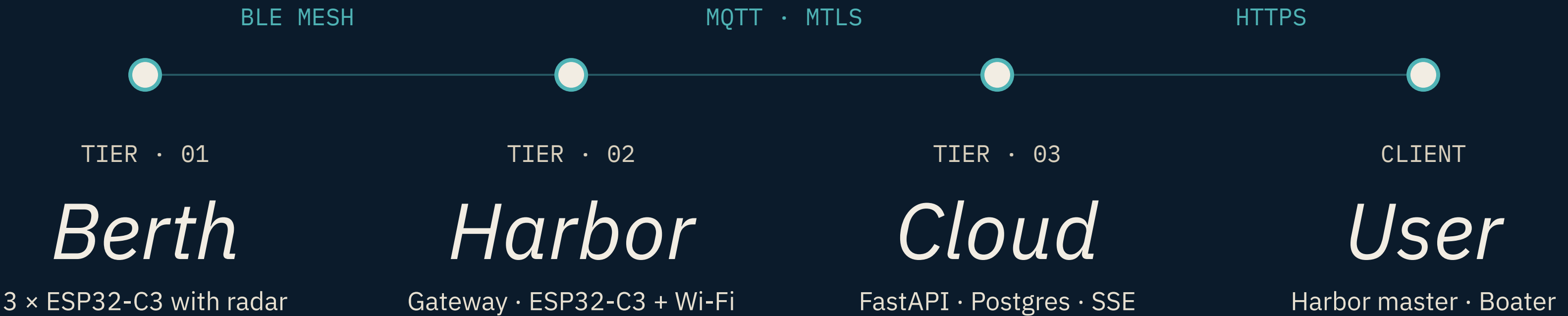
03

Dashboard.

1 FOR EVERYONE

React · live map · SSE push

From radar to dashboard.



Adopt with a *QR scan*. Replace by scanning the new one.

STEP 01

Scan QR

STEP 02

PB-ADV
handshake

STEP 03

Address
mapping

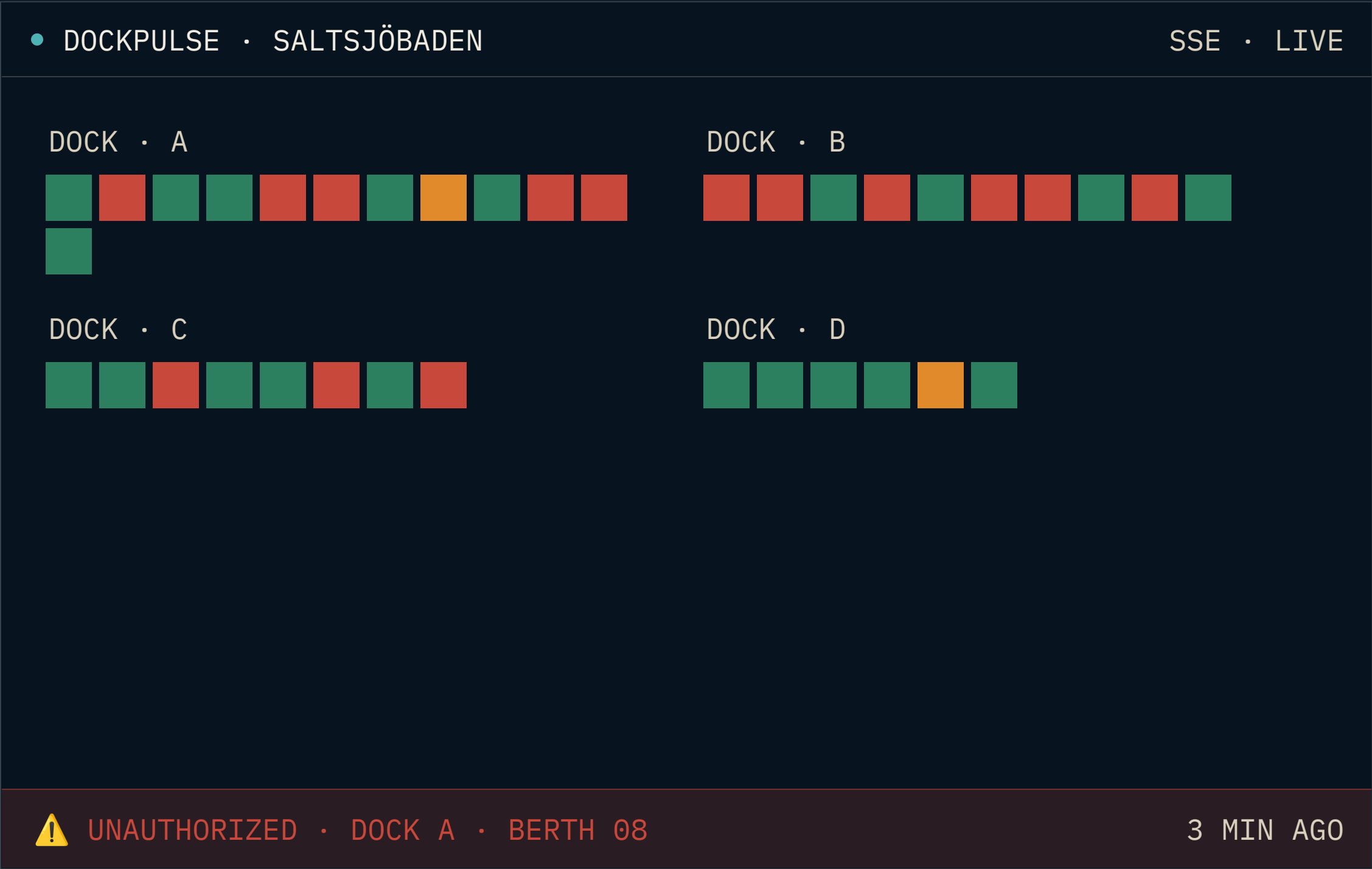
STEP 04

Live on the map



Hot-swap a failed sensor by rescanning.No reconfiguration.

One live map per harbor.



LEGEND

Free · tenant has declared availability

Occupied · vessel detected by radar

Visitor / unauthorized mooring

- VIEWS
- Color-coded harbor map (live)
- Adoption flow with QR scan
- Alert panel — unauthorized mooring
- Node-health view (battery, signal)
- Tenant page — set return date
- Email notifications via Resend

Every *Must, Should* — and the *Could*.

ID	FEATURE	WHAT WAS DELIVERED	PRIORITY
F1	Berth occupancy sensing	mmWave radar over UART, occupancy filter, raw diagnostic stream.	MUST
F2	Local status indicator	Dual-color LED, updates instantly on state change.	MUST
F3	Gateway communication	BLE Mesh ingest, MQTT/mTLS uplink at QoS 1, LWT lifecycle.	MUST
F4	Backend server	FastAPI + Postgres, REST + SSE, multi-tenant, Alembic.	MUST
F5	Web dashboard	React + TS, color-coded map, alert panel, node-health view.	MUST
F6	Unauthorized mooring detection	Dashboard alerts plus email via Resend.	SHOULD
F7	Node provisioning & hot-swap	QR → PB-ADV handshake → live on the map. Hot-swap by rescanning.	SHOULD
F8	Berth booking system	Mark availability, set return date, email confirmations.	COULD

What we descoped — and why.

ID	FEATURE	REASON	STATUS
F4.c	Multi-harbor map view	Backend is multi-tenant; cross-harbor switcher view descoped at planning.	DESCOPED · COULD
F5.c	Mesh-topology view	Diagnostic view of the BLE Mesh fabric. Descoped for operator-facing views.	DESCOPED · COULD
HW	Radar power-gating MOSFET	A power-gating MOSFET would enable deep sleep. Field test ran on mains.	OPEN · HARDWARE
QA	Multi-day dockside field test	Bench-tested end-to-end; a full diurnal cycle on a dock didn't fit the timeline.	OPEN · VALIDATION

What we'd keep — and change.

TECHNICAL

BIGGEST WIN

Contracts before code.

OpenAPI · JSON on MQTT · packed binary on the mesh wire.

BIGGEST REGRET

Bench, not field.

No quantitative latency or false-positive numbers from a real dock yet.

TEAM

WHAT WE 'D KEEP

A written team agreement.

Code review on every change · Conventional Commits · revised twice from retros.

WHAT WE 'D CHANGE

Surface *under-delivery* earlier.

Daily scrum caught blockers fast; it slow-walked detection of missed commitments.

From prototype to harbor-ready.

01

Close the power loop.

Design the radar power-gating circuit, verify deep-sleep current in hardware, run a multi-day solar-charged test for a real autonomy figure.

02

Deliver a reliability report.

Deploy in a real harbor. Tune the filter with the diagnostic stream the firmware already publishes. Publish false-positive and false-negative rates an operator can trust.

ON THE BENCH

Debounce-and-threshold filter against bench data.

IN THE FIELD

Two-week dockside deployment, full diurnal cycle.

IN THE PRODUCT

Multi-harbor view · mesh-topology view · booking.

↓ SWITCH THE BROWSER ↓

Live *demo*

→ dockpulse.xyz

WALK

Harbor map · live SSE updates

WALK

QR-scan adoption flow

WALK

Tenant: set return date

WALK

Unauthorized mooring alert